

Heiroglyphy: Recovering the Conceptual Geometry of Ancient Egyptian Through Vector Space Alignment

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Abstract

Ancient Egyptian has been translated by scholars for centuries, but translation is inherently lossy. When *nṯr* becomes “god,” the web of relationships that defined divinity for an Egyptian speaker—its proximity to kingship, offering, truth, and cosmic order—is compressed into a single modern word. We present a method for recovering this lost semantic structure using vector space alignment. By training word embeddings on 100,729 Ancient Egyptian sentences and aligning the resulting geometric space to English, we construct a bridge that preserves not just word meanings but the *relationships between meanings*. Across 15 experimental iterations, we achieve 32.35% Top-1 unsupervised cross-lingual retrieval accuracy—without a bilingual dictionary—and demonstrate that the aligned space reveals conceptual structures invisible to literal translation: that gold and divinity occupy the same semantic region, that silence is the defining condition of death, that the Egyptian serpent is divine rather than wise, and that the “earth mother” archetype is absent from the Egyptian conceptual landscape. The methodology is domain-agnostic: anywhere two signal spaces share latent structure, this approach can recover it.

1 Introduction

Imagine two cities, each with their own map. The streets are labeled in different languages, the neighborhoods have different names, and the scale is different. But the *shape* of the city—the way neighborhoods cluster together, the distance between the market and the harbor, the fact that the temple sits between the palace and the cemetery—is the same. If you could find the right rotation and scaling, you could overlay one map onto the other, and every landmark would align.

This is the core hypothesis of this work: that languages, when represented as geometric objects, share a common shape. Words that mean similar things cluster together in every language. “Water” sits near “river” and “flood” whether you write it in English or in Middle Egyptian. The challenge is to find the transformation that aligns one language’s shape onto another’s.

What makes this interesting for Ancient Egyptian is not the translation itself—Egyptologists have been translating hieroglyphic texts for two centuries. It is what the *geometry* reveals that translation cannot. A dictionary tells you that *nṯr* means “god.” But the embedding space tells you that *nṯr* sits near *nḃw* (gold), near *nswt* (king), near *māq.t* (truth/cosmic order)—and *far* from where an English speaker would place “god” relative to “gold.” The distance between “god” and “gold” in Egyptian is nearly zero. In English, it is vast. This difference is invisible to translation. It is visible to geometry.

The purpose of this project is to build that geometric lens—to align the embedding space of Ancient Egyptian to English with enough fidelity that the *relational structure* of the ancient language becomes readable. The accuracy metric (32.35% Top-1 retrieval) is a means to an end. The end is the discoveries: the conceptual architecture of a civilization that thought differently from us, preserved in the statistical structure of its texts.

2 Background: Words as Geometry

2.1 The Distributional Hypothesis

The foundation of this work is a simple observation from linguistics: words that appear in similar contexts tend to have similar meanings [Harris, 1954, Firth, 1957]. “Doctor” and “physician” appear near the same words (“patient,” “hospital,” “treat”), so they must mean similar things. This is the *distributional hypothesis*.

In the 2010s, researchers discovered that this hypothesis could be operationalized at scale [Mikolov et al., 2013a]. By training a neural network to predict a word from its context (or vice versa), each word in a language can be assigned a vector—a point in high-dimensional space—such that words with similar meanings are close together. The resulting space has remarkable properties: vector arithmetic encodes semantic relationships. The vector for “king” minus “man” plus “woman” yields a vector near “queen” [Mikolov et al., 2013b].

2.2 Cross-Lingual Alignment

A natural question follows: if every language forms a geometric shape, do different languages form the *same* shape? Research on cross-lingual word embeddings suggests they do [Conneau et al., 2018, Lample et al., 2018]. The word for “water” clusters near the word for “river” in every language. The overall topology is approximately isomorphic.

This means that if we can find a transformation—a rotation, scaling, and translation—that maps one language’s embedding space onto another’s, we can translate between them without a bilingual dictionary. The transformation can be learned from a small set of “anchor” pairs (words whose translation is known) and then applied to the entire vocabulary.

The most common approach is *Procrustes alignment* [Schönemann, 1966]: given anchor pairs (x_i, y_i) where x_i is a source-language vector and y_i is a target-language vector, find the linear transformation W that minimizes:

$$W^* = \arg \min_W \sum_i \|Wx_i - y_i\|^2 \quad (1)$$

This has a closed-form solution via singular value decomposition (SVD) when W is constrained to be orthogonal, or via Ridge regression when it is not.

2.3 The Ancient Egyptian Challenge

Applying cross-lingual alignment to Ancient Egyptian is harder than aligning, say, French and English:

- **No living speakers.** There is no bilingual community to validate results.
- **Sparse data.** The entire digitized corpus is $\sim 100,000$ sentences, with a median length of 6 words. Modern NLP models train on billions of sentences.
- **Domain bias.** The surviving texts are overwhelmingly funerary and religious. Everyday speech is almost entirely lost.
- **Morphological complexity.** Egyptian words are agglutinative, with suffix pronouns, plural markers, and editorial annotations embedded in the transliteration.

- **Semantic distance.** Ancient Egyptian is 4,000 years removed from Modern English. The conceptual categories are different. There is no guarantee that the embedding spaces are isomorphic.

Despite these challenges, the distributional hypothesis does not require a living language. It requires only a corpus large enough to estimate co-occurrence statistics. The BBAW hieroglyphic text database [BBAW, 2023] provides exactly that.

3 Methodology

3.1 Overview

The final pipeline, arrived at through 15 experimental iterations, consists of four stages:

1. Train 768-dimensional FastText embeddings on the Egyptian transliteration corpus.
2. Fuse text embeddings with 768-dimensional visual features (zero-padded) to create 1536-dimensional representations.
3. Learn a Ridge regression mapping from the fused Egyptian space to the 300-dimensional English GloVe space, using 5,360 anchor pairs for training.
4. Evaluate via nearest-neighbor retrieval on 1,340 held-out anchor pairs.

3.2 Data

Egyptian corpus. 100,729 sentences of transliterated hieroglyphic text from the Berlin-Brandenburg Academy’s Thesaurus Linguae Aegyptiae (TLA) [BBAW, 2023], accessed via the BBAW HuggingFace dataset. Total tokens: 789,159. Unique vocabulary: 80,662 (before filtering).

Anchor pairs. 8,541 Egyptian–English word pairs extracted via co-occurrence analysis from parallel German translations in the TLA corpus. Each pair has a confidence score (mean: 0.747, range: 0.30–1.00). After filtering for vocabulary coverage, 6,700 pairs are used (80/20 train/test split, `random_state=42`).

English embeddings. GloVe 6B 300d [Pennington et al., 2014]: 400,000 English words trained on 6 billion tokens of Wikipedia and Gigaword.

Visual features. 768-dimensional ResNet-50 features extracted from the HamdiJr/Egyptian_hieroglyphs dataset [HamdiJr, 2023]: 4,210 labeled hieroglyph images spanning 115 Gardiner sign classes. Visual match rate in the final model: 0.08%.

3.3 Egyptian Embeddings

We train FastText skip-gram embeddings [Bojanowski et al., 2017] on the Egyptian transliteration corpus. FastText is preferred over Word2Vec because its subword (character n-gram) architecture captures morphological structure—critical for Egyptian’s agglutinative morphology, where variants like *ntr*, *ntrw*, and *ntr.j* share a common root.

The final training configuration, selected through a 9-configuration parameter sweep (Section 4):

Parameter	Value
<code>vector_size</code>	768
<code>window</code>	10
<code>min_count</code>	5
<code>sg</code>	1 (skip-gram)
<code>epochs</code>	10

Table 1: FastText training parameters (V15 configuration).

Two parameters were critical discoveries:

`min_count=5` filters words appearing fewer than 5 times. The original model (`min_count=1`) retained 80,662 words, of which 52,813 (65.5%) were hapax legomena—words appearing exactly once. These single-occurrence words have essentially random vectors. Filtering to `min_count=5` reduces vocabulary to 10,833 words (an 87% reduction) while *improving* alignment accuracy. Less noise, better geometry.

`window=10` doubles the context window from the original 5. The Egyptian corpus has a median sentence length of 6 words. With `window=5`, most words see only 2–3 context neighbors. Widening to 10 allows FastText to capture relationships across the full span of typical sentences.

3.4 Fusion and Alignment

Each Egyptian word receives a 1536-dimensional fused representation:

$$\mathbf{v}_{\text{fused}} = [\mathbf{v}_{\text{text}} \parallel \mathbf{v}_{\text{visual}}] \quad (2)$$

where $\mathbf{v}_{\text{text}} \in \mathbb{R}^{768}$ is the FastText vector and $\mathbf{v}_{\text{visual}} \in \mathbb{R}^{768}$ is the ResNet-50 visual feature vector (or $\mathbf{0}$ if no visual mapping exists). In practice, 99.92% of words have zero visual features. The 1536-dimensional space nonetheless outperforms the 768-dimensional text-only space, likely due to implicit regularization from the zero-padding (Section 4).

The alignment is learned via Ridge regression:

$$W^* = \arg \min_W \|XW - Y\|_F^2 + \alpha \|W\|_F^2 \quad (3)$$

where $X \in \mathbb{R}^{n \times 1536}$ contains fused Egyptian vectors, $Y \in \mathbb{R}^{n \times 300}$ contains target GloVe vectors, and α is the regularization strength. The optimal $\alpha = 0.001$ was found through grid search evaluated by retrieval accuracy (not MSE—a critical distinction discussed in Section 4).

All projected Egyptian vectors are L2-normalized before retrieval.

3.5 Evaluation

For each test pair (e_i, g_i) , we project the Egyptian vector $\mathbf{v}_{e_i}$ to English space and retrieve its k nearest neighbors by cosine similarity in the full 400,000-word GloVe vocabulary. The pair is “correct” if g_i appears in the top- k results.

Metric	Score
Top-1 Accuracy	32.35%
Top-5 Accuracy	41.47%
Top-10 Accuracy	45.13%
Test pairs	1,340
Anchor coverage	78.4%
Visual match rate	0.08%

Table 2: Final evaluation metrics (V15).

4 The Iterative Journey

The final methodology emerged through 15 experimental iterations. Each failure was as informative as each success.

V	Technique	Top-1	Status
1	Neural Vec2Vec (Multi-Space)	—	Failed
2	Unsupervised Adversarial	—	Failed
3	Orthogonal Procrustes	22.0%	Baseline
4	Procrustes + CSLS	15.0%	Regression
5	Procrustes + 10× Data	24.5%	Improved
6	BERT Contextual	0.5%	Failed
7	FastText 768d	29.1%	Breakthrough
8	Coptic Bridge	28.2%	Regression
9	Text + Visual Fusion (1536d)	30.5%	First 30%+
10	Vocabulary Normalization	30.7%	Minor gain
11	MLP Neural Network	28.8%	Regression
12	Egyptian→German	12.9%	Exploratory
13	Ridge Alpha Tuning	31.6%	Improved
14	Iterative Procrustes + Hub Filter	31.6%	No change
15	FastText Retraining (mc5_w10)	32.4%	SOTA

Table 3: Accuracy progression across 15 iterations.

4.1 Key Lessons

Linear methods beat neural networks. With only $\sim 5,000$ training pairs, neural architectures (MLPs, GANs, BERT) consistently overfit. The Ridge regression closed-form solution outperforms all learned mappings. This is the central methodological result: for low-resource alignment, reach for linear algebra before deep learning.

BERT destroys ancient languages. The WordPiece tokenizer fragments hieroglyphic transliterations into meaningless subwords: *hr,w* (Horus) becomes [*h*, *##r*, *##*, *##w*]. Accuracy collapsed from 24.5% to 0.5%. Pre-trained NLP models encode modern linguistic assumptions that do not transfer to ancient languages.

Quality beats quantity for anchors. Adding 368 Coptic cognate anchors (V8) *reduced* accuracy. Etymology $\neq$ semantics: words change meaning over millennia. A smaller set of high-confidence anchors outperforms a larger set of mixed-quality ones.

MSE is anti-correlated with retrieval accuracy. Cross-validated Ridge regression selects $\alpha = 100$ by mean squared error. But the actual best retrieval accuracy occurs at $\alpha = 0.001$ —five orders of magnitude lower. Lower regularization produces “sharper” vectors that are noisier in an L2 sense but more discriminative for nearest-neighbor lookup. *Optimizing for regression loss optimizes against retrieval.*

Zero-padding helps via regularization. Concatenating 768 zeros (placeholder visual features) to each 768-dimensional text vector improves accuracy despite adding no information. The larger parameter space (1536×300 vs 768×300) acts as implicit regularization, preventing overfitting. This was confirmed by ablation: at matched regularization strength, 768d and 1536d perform identically.

65% of vocabulary is noise. Over half of the Egyptian vocabulary consists of words appearing exactly once (hapax legomena). Their vectors are random. Filtering them (`min_count=5`) reduces vocabulary by 87% and *increases* accuracy. The model was learning from noise it didn’t need to learn from.

60% of the test set is function words. The accuracy metric is depressed by function word ambiguity—many Egyptian function words (prepositions, pronouns) map to the same few English stopwords. On content words alone, the model performs substantially better. The “hubness problem” (82% of predictions being “the”) is not a model failure—it is the model correctly reflecting that function words dominate both languages.

5 Discoveries

The following findings emerge from probing the aligned V15 embedding space with semantic analogies, concept midpoints, and cross-domain cluster analysis. They represent what literal translation cannot capture: the relational structure of how Egyptians connected ideas.

5.1 Gold Is Divine Flesh

The midpoint of “gold” and “divine” in English, projected into the Egyptian space, finds *ntri* (divine, similarity 0.642) and *nbw* (gold, similarity 0.617) as the top results. They are geometrically colocated.

Modern readers treat the Egyptian statement “the flesh of the gods is gold” as poetic metaphor. The embedding space suggests otherwise. Gold and divinity are not *compared* in the Egyptian textual record—they are *colocated*. The same linguistic contexts that invoke divinity invoke gold, and vice versa. The model cannot distinguish them because the corpus does not distinguish them. This is not metaphor. It is ontology.

5.2 Silence Is the Condition of the Dead

The midpoint of “silence” and “death” returns variants of *m(w)t* (die/dead) as all five nearest neighbors. There is no Egyptian word between silence and death—they occupy the same point in semantic space.

The Egyptians called the necropolis *tā-sgr*—“the silent land.” The dead were *sgr.w*—“the silent ones.” The embedding space confirms that this was not euphemism. Silence was death’s defining quality. What the dead lost was not life in some abstract sense. It was voice.

5.3 Seeing Was an Act of Magical Power

The midpoint of “eye” and “knowledge” finds *jr,t* (eyes) at the top, then *ḥkā* (heka/magic) as the third-nearest word. Egyptian sight sits between knowing and spellcasting.

The Eye of Horus (*wḏā,t*) was simultaneously an organ of perception, a protective amulet, and a unit of mathematical measurement. Seeing, in the Egyptian worldview, was not passive reception. It was an active, potent act—closer to divination than observation. The embedding space captures this without being told.

5.4 The Snake Is Divine, Not Wise

The midpoint of “snake” and “wisdom” finds gods and divinity—not wisdom, knowledge, or cunning. Every top result is a *ntr* variant.

This directly contradicts the Greek-influenced reading of serpent symbolism. In the Greek tradition, the snake is associated with wisdom (the serpent of Asclepius, the caduceus). In the Egyptian tradition, the serpent—the uraeus cobra on the pharaoh’s brow, the Apophis serpent of chaos—is associated with *divine power*. The embedding space separates these cultural readings without any awareness of Greek mythology.

5.5 Temple Is to House as God Is to Man

The vector analogy *house : temple :: man : ?* returns *ntr* (god). The proportional relationship is preserved across the 4,000-year alignment: a temple is a god’s house, in the same geometric sense that a house is a man’s dwelling.

This is one of the few cases where vector arithmetic—the technique that produces “king – man + woman = queen” in English—works across an ancient language boundary.

5.6 Mother Is Royalty, Not Earth

The midpoint of “mother” and “earth” does not find earth, soil, or land. It finds *hm,t-nzw* (royal wife) and *zā,t-nzw* (king’s daughter). In the Egyptian textual record, motherhood is a *royal* concept.

The “earth mother”—the fertile, nurturing ground-goddess—is an Indo-European archetype (Gaia, Terra, Demeter). The Egyptian mother-goddess (Isis, Hathor) is not earthy. She is regal. The embedding space reveals this cultural divergence without being told about either tradition.

5.7 Truth and Power Are the Same Force

The midpoint of “truth” and “power” finds *shm* (power/authority) at #1 and *hft.pl* (enemies) at #4. Truth, power, and the defeat of enemies occupy the same semantic region.

This reflects *māq.t*—usually translated as “truth” or “justice” but encompassing cosmic order, righteous authority, and the subduing of chaos. Truth in the Egyptian worldview is not passive correctness. It is the active force that maintains the universe against its enemies.

5.8 Love and Fear Meet at Eternity

The midpoint of “love” and “fear” finds *r-(n)hh* (eternal/forever) as the second-nearest word. Between love and fear, the Egyptians placed eternity.

This may reflect the theology of divine favor: the gods’ love is awe-inspiring, their wrath is terrifying, and both extend beyond time. The offering formula—“that he may be loved and feared for eternity”—collapses these concepts into a single liturgical act.

6 Discussion

6.1 Limitations

The primary limitation is corpus bias. The surviving Egyptian texts are overwhelmingly funerary and religious. The embedding space reflects the language of temples, tombs, and offering formulas—not the language of markets, households, or daily conversation. Discoveries about “the Egyptian worldview” are more precisely about the *literate priestly worldview preserved in stone*.

The 32.35% Top-1 accuracy, while significant for unsupervised cross-lingual alignment of a dead language, means that two-thirds of test words do not retrieve their correct translation. The geometric relationships we analyze are *statistical tendencies*, not certainties.

Additionally, the anchor pairs are extracted from German translations, introducing a second layer of cultural mediation. An Egyptian→German→English pipeline may introduce biases that a direct Egyptian→English approach would avoid.

6.2 What Translation Misses

The core contribution of this work is not the translations—Egyptologists already have those—but the *distances*. A dictionary can tell you that *ntr* means “god” and *nbw* means “gold.” But only the embedding space can tell you that they are essentially the same word, distributionally speaking. This is the kind of information that translation destroys: the proximity, the neighborhood, the geometric context.

Every act of translation is a projection from a high-dimensional space to a lower one. Meaning is lost. The embedding approach preserves more of the original dimensionality, allowing us to see relationships that the projection to English text made invisible.

6.3 Generalizability

The methodology is domain-agnostic. The question “can we align two signal spaces that share latent structure?” applies wherever:

- Two systems produce high-dimensional representations of related phenomena
- A small number of anchor correspondences can be identified
- The underlying structure is approximately isomorphic

Ancient Egyptian is a particularly compelling test case because it maximizes the difficulty: the languages are separated by 4,000 years, the corpus is small, and the conceptual categories are genuinely different. If vec2vec alignment works here, it can work in less extreme settings.

7 Conclusion

We set out to build a lens—a way of reading the Ancient Egyptian textual record that reveals what literal translation leaves behind. The lens is imperfect. It resolves some features clearly and blurs others. But where it does resolve, it shows us a world organized differently from our own: a world where gold *is* divinity, where silence *is* death, where the snake is sacred rather than wise, where truth is a weapon and motherhood is a crown.

These discoveries emerged not from sophisticated neural architectures or massive compute, but from the simplest possible alignment: a linear projection learned from a few thousand anchor pairs. The central technical lesson is that for low-resource cross-lingual alignment, simplicity is not a compromise—it is the optimal strategy. The central humanistic lesson is that the geometry of a language encodes its worldview, and that worldview is recoverable even after 4,000 years of silence.

Translation gave us the words. The vectors gave us the world between them.

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